

Test Case ID	Test Scenario /Description	Testing Type	Testing Strategy	Test Data / Input	Expected Result	Actual Result	Status
01	Register new user with valid information	Unit / Integration	Black Box	Name: Kavindya, Email: user@gmail.com, Password: User@123	“Signin successfully” Message should appear	As Expected	Pass
02	Register using existing email address	Unit / Integration	Black Box	Existing email: user@gmail.com	System should display "User already exists"	As Expected	Pass
03	Register with invalid pw	Unit	Black Box	Password: user123	Validation message "Invalid login credentials" should appear	As Expected	Pass
04	Register with empty required fields	Unit	Black Box	Blank form submission	Required field validation warnings should display	As Expected	Pass
05	Verify registration data saved in Supabase Auth	Integration	White Box	Valid registration request	User credentials should be inserted into Supabase authentication table	As Expected	Pass

06	Login with valid registered credentials	Unit / Integration	Black Box	user@gmail.com User@123	User should login successfully and redirect to homepage	As Expected	Pass
07	Login with unregistered email	Integration	Black Box	User@gmail.com User@123	"invalid login credentials" error should display	As Expected	Pass
08	Login using Google account authentication	Integration	Black Box	Select Google account	User should authenticate successfully through Google OAuth	As Expected	Pass
09	Click logout button after login	Integration	Black Box	Logout action	Session should clear and user redirected to login page	As Expected	Pass
10	Access protected dashboard without login	Integration	Black Box / White Box	Direct URL access to /dashboard	System should redirect unauthenticated user to login page	As Expected	Pass
11	Verify middleware checks session token on protected routes	Integration	White Box	Access route with/without session	Middleware should allow authorized users and block unauthorized users	As Expected	Pass

12	Verify successful login redirects user properly	White Box / Integration	White Box	Valid login request	System should route user to homepage automatically	As Expected	Pass
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PRODUCT MANAGEMENT MODULE

Test Case ID	Test Scenario/ Description	Testing Type	Testing Strategy	Test Data / Input	Expected Result	Actual Result	Status
01	Add a product with all valid details	Unit / Integration	Black Box	Name: Dell XPS 15, Price: 350000, Stock: 5, Brand : Apple	Product should be created successfully and displayed in product list	As Expected	Pass
02	Submit product with empty product name	Unit	Black Box	Product name: brand: price: stock:	Validation warning "Product name required" should display	As Expected	Pass
03	Submit product with invalid negative price	Unit	Black Box	Price: - 1000	System should show error as price must be positive number	As Expected	Pass

05	Submit product without selecting category	Unit	Black Box	Category blank	Category required validation message should display	As Expected	Pass
06	Upload single valid image file	Integration	Black Box	jpg/png image	Successfully uploaded image	As Expected	Pass
11	Verify product data inserted into Supabase products table	Integration	White Box	Valid add request	New product row should be created in products table	As Expected	Pass
12	Update product price and stock	Integration	Black Box	Change price to 340000, stock to 4	Updated values should save and display correctly	As Expected	Pass
13	Modify product specifications	Integration	Black Box	Change RAM to 32GB	New specification should update successfully	"cannot coerce the result to a single JSON object"	unsure
14	Verify Supabase update query	Integration	White Box	Edit request	Existing product row should be	As Expected	Pass

	executes correctly				updated in database		
15	Delete selected product from admin	Integration	Black Box	Click delete product	Product should be removed from product list	As Expected	pass
16	Verify deleted product removed from Supabase table	Integration	White Box	Delete request	Product row should be deleted permanently from DB	As Expected	Pass
17	Delete product containing uploaded images	White Box / Integration	White Box	Product with 3 images	Associated images should also be removed from storage if cleanup enabled	As Expected	Pass
20	Add product edit product delete product	integration	Black Box / White Box	Change product details and then click save changes	Should change product details or deleted product	As Expected	pass

AUCTION MANAGEMENT MODULE

Test Case ID	Test Scenario / Description	Testing Type	Testing Strategy	Test Data / Input	Expected Result	Actual Result	Status
01	Create auction with valid product	Unit / Integration	Black Box	Product: Dell XPS, Price: 200000	Auction should be created successfully and displayed in auction list	As Expected	Pass
02	Create auction without selecting product	Unit	Black Box	Product blank	Validation warning should appear for name, Brand	As Expected	Pass
03	Enter reserve price as zero	Unit	Black Box	Reserve Price: 0	System should reject invalid reserve price	Create auction product	Ensure(it is optional)
04	Enter negative reserve price	Unit	Black Box	Reserve Price: -5000	Validation error should display	Product created successfully with alert	Ensure(because it is optional)

05	Select auction end date earlier than start date	Unit	Black Box	Start: 5 May, End: 4 May	System should reject invalid auction duration	As Expected	Pass
06	Verify auction details stored in Supabase database	Integration	White Box	Valid create auction request	Auction row should be inserted into auctions table	As Expected	Pass
07	Verify countdown timer starts after auction activation	Unit / Integration	White Box	5 minute auction	Timer should decrease every second accurately	As Expected	Pass
08	End auction with bids above reserve price	Integration	White Box	Highest bid exists	Highest bidder should be marked as auction winner	As Expected	Pass
09	Cancel auction before completion	Integration	Black Box	Click Cancel Auction	should update product successfully	As Expected	Pass
10	Verify status update query after	Integration	White Box	Status transiti	Auction status should	As Expected	Pass

	start/end/cancel			on request	correctly update in database		
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REAL-TIME BIDDING SYSTEM

Test Case ID	Test Scenario / Description	Testing Type	Testing Strategy	Test Data / Input	Expected Result	Actual Result	Status
01	Place first valid bid on active auction	Unit / Integration / System	Black Box	Current price: 100000, User bid: 1001001	Bid should be placed successfully with alert	As Expected	Pass
02	Submit bid lower than current highest bid	Unit	Black Box	Current highest: 20000, New bid: 1001001	Should show error message "Bid must be higher than current bid of 20000"	As Expected	Pass
03	Submit bid equal to current highest bid	Unit	Black Box	Current highest: 20000, New bid: 1001001	Should show error message "Bid must be higher than current bid of 20000"	As Expected	Pass
05	Submit negative bid amount	Unit	Black Box	-500000	Should show error message "Bid must be higher than current bid of 20000"	As Expected	Pass
06	Submit higher bid than	Integration	Black Box	Current highest: 110000,	Should bid placed successfully with alert	As Expected	Pass

	current highest			New bid: 120000			
07	Verify valid bid stored in Supabase bids table	Integration	White Box	Valid bid request	New bid row should be inserted in bids table	As Expected	Pass
08	Refresh page after bid placed	integration	Black Box	Browser refresh	Latest highest bid should still display from database	As Expected	Pass

SHOPPING CART MODULE

Test Case ID	Test Scenario / Description	Testing Type	Testing Strategy	Test Data / Input	Expected Result	Actual Result	Status
01	Add a product to empty cart	Unit / Integration	Black Box	Click add to cart in Dell Inspiron 5620-i5	Product should appear in cart successfully with alert	As Expected	Pass
02	Add multiple different products	Integration	Black Box	Click add to cart in Dell Inspiron 5620-i5 & Asus VivoBook F1504VAP -Core 5	Both items should display in cart list	As Expected	Pass

03	Add same product again	Unit / Integration	Black Box	Add Dell XPS twice	Quantity should increase	As Expected	Pass
04	Increase quantity using + button	Unit	Black Box	Click + mark	Quantity should update correctly	As Expected	Pass
05	Decrease quantity using - button	Unit	Black Box	Click - mark	Quantity should reduce correctly	As Expected	Pass
06	Remove selected product from cart	Integration	Black Box	Click Remove	Product should disappear from cart	As Expected	Pass
07	Add items and inspect browser local storage	Integration	White Box	Add products	Cart JSON data should save in local storage	As Expected	Pass
08	Display total amount	Unit	Black Box	Display total amount with subtotal and tax	Should Display total amount correctly	As Expected	Pass

6. Order / Checkout Module

Test Case ID	Test Scenario / Description	Testing Type	Testing Strategy	Test Data / Input	Expected Result	Actual Result	Status
01	Submit order with all valid customer details	Unit / Integration	Black Box	Name, Email Address, Phone number, Shipping Address, City, postal code	Order should be placed successfully	As Expected	Pass
02	Submit order with empty address field	Unit	Black Box	Shopping Address blank	Validation warning should display like please fill in this field	As Expected	Pass
03	Verify details transferred to order summary	Integration	Black Box	Valid checkout details	Order summary should show correct shipping information	As Expected	Pass
04	Verify new order inserted into Supabase orders table	Integration	White Box	Submit order request	New order row should be created in database	As Expected	Pass

05	Admin changes order status Pending → Delivered	Integration	Black Box / White Box	Update status	Order status should save and display correctly	As Expected	Pass
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ADMIN DASHBOARD MODULE

Test Case ID	Test Scenario / Description	Testing Type	Testing Strategy	Test Data / Input	Expected Result	Actual Result	Status
DASH_001	Verify total revenue amount shown from completed orders	Integration	Black Box	Existing delivered orders	Correct total revenue should display	As Expected	Pass
DASH_002	Verify number of live auctions displayed	Integration	Black Box	Active auctions in DB	Correct active auction count should show	As Expected	Pass
DASH_003	Verify pending order quantity shown	Integration	Black Box	Pending orders exist	Correct pending order count should display	As Expected	Pass

DASH_004	Verify registered total user count display	Integration	Black Box	Users table records	Correct user count should show	As Expected	Pass
DASH_005	Verify Supabase count/sum queries execute correctly	Integration	White Box	Fetch dashboard stats	Correct aggregate values should return from DB	As Expected	Pass

SECURITY & ACCESS CONTROL MODULE

Test Case ID	Test Scenario / Description	Testing Type	Testing Strategy	Test Data / Input	Expected Result	Actual Result	Status
01	Access protected dashboard URL without login	Integration	Black Box	Direct URL: /dashboard	Error message like 404 This page could not be found	As Expected	Pass
02	Normal user tries accessing /admin page	Integration	Black Box	Buyer or seller account access	Access should be denied with error msg like 404	As Expected	Pass
03	User attempts product delete action	Integration	Black Box	User clicks delete	System should block with error message like "a	As Expected	Pass

					server error occur”		
SEC_004	Access protected page with expired/no session	Integration	White Box	Invalid session token	Middleware should reject access	As Expected	Pass
SEC_005	Unauthorized user attempts direct restricted DB query	Integration	White Box	Restricted query request	Supabase RLS should deny data access	As Expected	Pass
SEC_006	Logout and try browser back/protected URL	integration	Black Box / White Box	Logout performed	System should block with error message like “a server error occur”	As Expected	Pass